FINITE GROUPS THAT ARE THE PRODUCT OF EVERY PAIR OF NON-CONJUGATE MAXIMAL SUBGROUPS ARE SOLUBLE

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Abstract. We prove that every finite group that is the product of every pair of its non-conjugate maximal subgroups is soluble, answering Problem 10.34 of the Kourovka Notebook. The almost-simple case was proved by Tikhonenko and Tyutyanov. For the remaining minimal-counterexample branch, with unique minimal normal subgroup S^k and $k \geq 2$, two automorphism-stable coordinate subgroup classes satisfying the supplement criterion yield non-conjugate maximal supplements. The factorization hypothesis would then impose a p-adic demand growing linearly with k, whereas the wreath-product quotient contributes only coordinate outer automorphisms and $k!$. One fixed valuation gap therefore rules out this branch for every $k \geq 2$. Using the classification of finite simple groups (CFSG) and published subgroup and order data, uniform parabolic, torus, and primitive-prime-divisor arguments cover the infinite families; stable flag parabolics handle graph fusion; GAP certificates cover the designated finite and sporadic cases.

1. Introduction

This section states the problem and answer, identifies the new mechanism and its inherited inputs, and gives a concise roadmap of the proof and its verification boundary.

Problem 10.34 of the Kourovka Notebook, posed by V. S. Monakhov in 1986 [18], asks:

> *Does there exist a finite non-soluble group that coincides with the product of any two of its non-conjugate maximal subgroups?*

Many soluble groups have this property: it holds vacuously whenever all maximal subgroups are conjugate, and $S_4 = AB$ for every pair of non-conjugate maximal subgroups A, B. The question is whether this factorization condition forces solubility. It does.

Theorem 1.1 (Main theorem)**.** *Let G be a finite group such that $G = AB$ for every pair of maximal subgroups $A, B \leq G$ that are not conjugate in G. Then G is soluble.*

We say that G has *property* P when it satisfies the hypothesis of Theorem 1.1. Tikhonenko and Tyutyanov proved the almost-simple case [34]; their result supplies the $k = 1$ branch of our proof. They also record that Zenkov had announced a negative answer in a 1997 conference abstract [38], while stating that no proof of the announcement was known to them. The July 2026 Kourovka Notebook still lists the problem as unsolved [18]. Accordingly, this paper proves the theorem but does not claim priority for the conclusion. After this paper's arXiv v1 and v2, dated 4 August 2026 and 6 August 2026, respectively, Li and Yang submitted a separately authored preprint dated 19 August 2026 that states the same theorem [22]. Their proof uses product–socle lifting from nonfactorizing almost-simple coordinate subgroups rather than the stable-class valuation obstruction. These are distinct proof architectures.

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For comparison, Theorem 4.4 also recovers the almost-simple conclusion; it is not used in the proof of Theorem 1.1.

The maximal factorizations of finite simple groups and their automorphism groups were classified by Liebeck, Praeger, and Saxl [21]. Those tables identify the factorizations that occur. Here the decisive step is instead a reusable obstruction: two local subgroup classes in a simple coordinate group force an impossible divisibility condition in every transitive direct-power extension. The classification of finite simple groups, maximality and order data, automorphism descriptions, parabolic theory, and Zsigmondy's theorem remain external inputs to the family coverage.

Proof mechanism. Suppose that a minimal counterexample G exists. Its unique minimal normal subgroup has the form $N = S^k$, where S is non-abelian simple, and G has a transitive coordinate realization inside an automorphism wreath product. If $[V]$ is X-stable and self-normalizing, maximal in the stable-class poset $\mathcal{P}_X(S)$, and normally saturating, then it produces a maximal supplement $B_V = N_G(V^k)$ whose order is $|G/N||V|^k$.

If $[V]$ and $[W]$ are distinct suitable classes, property P forces $G = B_V B_W$. The subgroup-product formula then imposes a p-adic divisibility condition whose exponent grows linearly with k. By contrast, the p-part available in G/N is bounded by the coordinate outer automorphisms and $k!$. A fixed valuation gap therefore gives a contradiction for every $k \geq 2$.

It remains to construct one such pair for each non-abelian finite simple group. Uniform parabolic, torus, and primitive-prime-divisor arguments handle the infinite families. Graph automorphisms require stable flag-parabolic classes in several families. GAP certificates handle the designated finite base, sporadic groups, and exceptional parameters.

Contribution and dependence. The conceptual hierarchy of the proof is summarized below. Imported classification and primitive-prime-divisor results are inputs, not claims of novelty.

Component	Status
Almost-simple $k = 1$ result	Prior theorem of Tikhonenko–Tyutyanov
CFSG and maximal-subgroup/order data	Cited external inputs
Reduction to a unique socle S^k	Standard ingredients assembled for property P
Suitable stable, self-normalizing classes satisfying poset maximality and normal saturation	Central maximal-supplement machinery with exact order
Uniform-in-k product-supplement valuation obstruction	Principal conceptual result
Stable flag parabolics under graph fusion	Distinctive structural mechanism
Infinite-family coverage	Application of the criterion using cited classification and arithmetic inputs
Finite and sporadic coverage	GAP-certified component
Lean and Rocq developments	Partial, conditional formal verification; not an end-to-end proof

TABLE 1. New, inherited, and externally supplied components.

The mathematical yield is the passage from local stable-class data in S to an obstruction valid for every multiplicity k. Poset maximality and normal saturation are the structural assumptions that turn local classes into maximal supplements; the valuation gap is the transferable numerical input. The graph-fusion cases show that ordinary maximal subgroups are not essential: stable flag parabolics can replace fused maximal classes.

The manuscript proves the theorem by ordinary mathematical argument. Lean checks named structural and arithmetic statements under explicit hypotheses; Rocq/MathComp supplies a direct-power component through an audited interface; GAP certifies designated finite cases; and Python checks manifests, certificates, source maps, and independent arithmetic. Appendix C states exactly how these components compose and which claims remain external.

2. Reduction to the monolithic coordinate case

By the end of this section, every counterexample has been converted into a transitive S^k-coordinate problem. The remaining task will be to construct two incompatible maximal supplements from local subgroup classes of S. The quotient argument and the monolithic reduction are standard; the explicit coordinate closure records exactly the arithmetic contribution available above the socle.

Lemma 2.1 (Factorizations and conjugate representatives)**.** *If $G = AB$ for subgroups A, B, then $G = A^x B^y$ for all $x, y \in G$.*

Proof. Since $G = AB$, write $xy^{-1} = ab$ with $a \in A$, $b \in B$. Then, as subsets of G,

$$A^x B^y = x^{-1}Ax\, y^{-1}By = x^{-1}A(ab)By = x^{-1}(AB)\, y = x^{-1}Gy = G. \qquad \square$$

Consequently, property P depends only on the conjugacy classes of maximal subgroups: it says every pair of *distinct classes* has one (hence every) representative pair factorizing G.

Lemma 2.2. *Property* P *is inherited by quotients.*

Proof. Maximal subgroups of G/K are the images of the maximal subgroups of G containing K; non-conjugate ones lift to non-conjugate ones, and factorizations project. □

Proposition 2.3 (Minimal counterexample structure)**.** *Let G have least order among the non-soluble groups with property* P*. Then G has a unique minimal normal subgroup $N = S^k$ with S non-abelian simple and $k \geq 1$, $C_G(N) = 1$, G/N is soluble, and G embeds in $\mathrm{Aut}(S) \wr S_k$ with N mapping onto $\mathrm{Inn}(S)^k$ and G acting transitively on the k coordinates.*

Proof. Every proper quotient of G has property P (Lemma 2.2), hence is soluble by minimality of $|G|$. The soluble radical R of G is trivial: otherwise G/R is non-soluble (as R is soluble) with property P, contradicting minimality. So the socle of G is a direct product of non-abelian minimal normal subgroups. If $N_1 \neq N_2$ were two of them, then G/N_1 is soluble while $N_2 \cong N_2N_1/N_1 \leq G/N_1$ is non-soluble, a contradiction. Hence there is a unique minimal normal subgroup $N = S^k$, S non-abelian simple. Now $C_G(N) \cap N = Z(N) = 1$ since N is a direct product of non-abelian simple groups; if $C_G(N)$ were nontrivial it would contain a minimal normal subgroup of G, necessarily N by uniqueness, contradicting $C_G(N) \cap N = 1$. So $C_G(N) = 1$, and the conjugation action embeds G into $\mathrm{Aut}(N) = \mathrm{Aut}(S) \wr S_k$ in the standard way; minimal normality of N forces transitivity on coordinates. Published references for the two standard structural inputs used here are [39, Lemma 2.10, pp. 173–174] for the automorphism group of a direct power and [23, p. 432] for the resulting transitive wreath realization in the unique-minimal-normal-subgroup setting. Finally, G/N is soluble because it is a proper quotient. □

Definition 2.4 (Coordinate closure)**.** For G as in Proposition 2.3 let G_1 be the stabilizer of the first coordinate and $\pi_1 \colon G_1 \to \mathrm{Aut}(S)$ the projection. Set $X := \pi_1(G_1) \cdot \mathrm{Inn}(S)$, the *coordinate closure*. By coordinate-transitivity all coordinates give the same X up to conjugacy, and $\mathrm{Inn}(S) \leq X \leq \mathrm{Aut}(S)$.

Lemma 2.5 (Coordinate normalization)**.** *With X as in Definition 2.4, the wreath realization can be conjugated inside* $\operatorname{Aut}(S) \wr S_k$ *so that*

$$N \le G \le X \wr S_k.$$

Every component automorphism of every element of the conjugated copy of G then lies in X.

Proof. Use the left-conjugation maps $c_g(z) = gzg^{-1}$, while retaining $A^g = g^{-1}Ag$ for subgroup conjugates. If an element sends S_i to S_j, its $i \to j$ component is the induced automorphism under the fixed identifications $S_i \cong S \cong S_j$; compositions of these automorphisms are read in the ordinary right-to-left order. Fix $g_j \in G$ carrying coordinate 1 to coordinate j, with $g_1 = 1$, and let $a_j \in \operatorname{Aut}(S)$ be its $1 \to j$ component. If $g \in G$ carries coordinate i to coordinate j through the component $c \in \operatorname{Aut}(S)$, then $g_j^{-1}\, g\, g_i$ stabilizes coordinate 1, because its left-conjugation map follows

$$S_1 \xrightarrow{c_{g_i}} S_i \xrightarrow{c_g} S_j \xrightarrow{c_{g_j^{-1}}} S_1.$$

Its component is $a_j^{-1} \circ c \circ a_i$, abbreviated $a_j^{-1} c a_i$, and therefore lies in $\pi_1(G_1) \le X$. Put $\delta := (a_1, \dots, a_k) \in \operatorname{Aut}(S)^k$. For an element with $i \to j$ component c, the corresponding component after replacing G by $\delta^{-1} G \delta$ is exactly $a_j^{-1} c a_i \in X$. Thus every component of every element of the conjugated group lies in X, while $a_1 = 1$ leaves X itself unchanged. This is the required normalized realization. □

Convention 2.6 (Normalized coordinate model). Henceforth G denotes the conjugate supplied by Lemma 2.5. Property P and all criteria below are invariant under this replacement. They are also invariant under replacing G by a conjugate under $(a, \dots, a) \in \operatorname{Aut}(S)^k$, so statements may be proved after normalizing X up to $\operatorname{Aut}(S)$-conjugacy.

Convention 2.7 (Coordinate action and arithmetic notation). For *coordinate* calculations we fix the following convention. Write $c_g(z) := gzg^{-1}$ (left conjugation), let S_i be the i-th coordinate factor of $N \cong S^k$, and let $\sigma_g \in S_k$ be the permutation determined by $c_g(S_j) = S_{\sigma_g(j)}$. There are component automorphisms $a_{g,i} \in \operatorname{Aut}(S)$ such that

$$\big(c_g(z)\big)_i = a_{g,i}\big(z_{\sigma_g^{-1}(i)}\big) \qquad (z \in N,\ 1 \le i \le k):$$

the value arriving at coordinate i comes from the *inverse-permuted* source coordinate $\sigma_g^{-1}(i)$. From $c_{gh} = c_g \circ c_h$ these data compose by

$$\sigma_{gh} = \sigma_g \sigma_h, \qquad a_{gh,i} = a_{g,i}\, a_{h,\sigma_g^{-1}(i)}.$$

After the normalization of Convention 2.6, every $a_{g,i}$ lies in X. Throughout, c_g denotes left conjugation on elements, whereas $A^g = g^{-1}Ag$ denotes conjugation of subgroups; thus $A^g = c_{g^{-1}}(A)$. Write

$$x := |X/\operatorname{Inn}(S)|, \qquad t := |G/N|.$$

Lemma 2.8 (Wreath-top quotient divisor)**.** *With the normalized coordinate model and notation above,*

$$t \mid x^k k!.$$

Proof. The quotient map $X \to X/\operatorname{Inn}(S)$ induces

$$\theta\colon X \wr S_k \longrightarrow (X/\operatorname{Inn}(S)) \wr S_k$$

with kernel $\operatorname{Inn}(S)^k$. Under the conjugation embedding of Proposition 2.3, the subgroup $N = S^k$ acts as $\operatorname{Inn}(S)^k$, so $N \le G \cap \ker\theta$. Conversely, if $g \in G \cap \operatorname{Inn}(S)^k$, choose $n \in N$ inducing the same tuple of inner automorphisms on N. Then $gn^{-1} \in C_G(N) = 1$ by Proposition 2.3, and hence $g = n$. Thus $G \cap \ker\theta = N$, and G/N embeds in $(X/\operatorname{Inn}(S)) \wr S_k$. Lagrange's theorem gives

$$|G/N| \mid |(X/\operatorname{Inn}(S)) \wr S_k| = x^k k!,$$

as required. □

The following observation is not needed for the proof of Theorem 1.1 but explains why the pairs used later are of "supplement" type.

Remark 2.9 (Top–supplement pairs always factorize). If $M \supseteq N$ is maximal and B is maximal with $B \not\supseteq N$, then $BN = G$ (BN is a subgroup strictly containing the maximal B), and since $N \le M$ we get $MB \supseteq NB = G$. Hence G has property P if and only if G/N has property P and the product of every pair of non-conjugate maximal subgroups not containing N is G.

3. Stable classes and maximal product supplements

This section identifies the precise hypotheses under which a local class in S yields a maximal subgroup of every admissible S^k-extension. Stability and self-normalization give the supplement, intersection, and exact order; poset maximality and normal saturation give maximality. The nonconjugacy statement and graph-fusion substitute complete the engine recorded in Corollary 3.7.

Fix G as in Conventions 2.6 and 2.7: $N = S^k \le G \le X \wr S_k$, $t = |G/N|$, $x = |X/\operatorname{Inn}(S)|$. For $V \le S$ write $[V]$ for its S-conjugacy class. An S-class is X*-stable* if it is invariant under the action of X, i.e. V^a is S-conjugate to V for every $a \in X$ (Inn-stability being automatic).

Definition 3.1. $\mathcal{P}_X(S) := \{\, [V] : 1 < V < S,\ N_S(V) = V,\ [V]\ X\text{-stable} \,\}$, partially ordered by containment up to S-conjugacy ($[V] \le [U]$ iff $V \le U^s$ for some $s \in S$). Call V *normally saturating* if $\langle V^W \rangle = W$ for every overgroup $V \le W \le S$.

Remark 3.2. If V is a *maximal* subgroup of S and $[V]$ is X-stable, then $[V] \in \mathcal{P}_X(S)$ (maximal subgroups of a non-abelian simple group are self-normalizing), $[V]$ is a maximal element of the poset, and V is normally saturating (its only overgroups are V and S, and $\langle V^S \rangle = S$ by simplicity). Thus the lemmas below apply to maximal classes with no further hypotheses.

Lemma 3.3 (Product-supplement construction)**.** *Let* $[V] \in \mathcal{P}_X(S)$ *and set* $B_V := N_G(V^k)$. *Then*

(1) $B_V \cap N = N_S(V)^k = V^k$*;*
(2) $B_V N = G$*;*
(3) $|B_V| = t \cdot |V|^k$.

Proof. (1) An element of $N = S^k$ normalizes V^k iff each coordinate normalizes V. (2) Frattini-type argument: let $g \in G$. In the coordinate-action notation of Convention 2.7, $(V^k)^g = c_{g^{-1}}(V^k)$, and since every coordinate of V^k is V,

$$(V^k)^g = \prod_{i=1}^{k} a_{g^{-1},i}(V),$$

each factor is the image of V under a component automorphism $a_{g^{-1},i} \in X$. By X-stability there are $s_i \in S$ with $a_{g^{-1},i}(V) = V^{s_i}$. So $(V^k)^g$ is N-conjugate to V^k: there is $n \in N$ with $(V^k)^{gn} = V^k$, i.e. $gn \in B_V$, so $g \in B_V N$. (3) $|B_V| = |B_V N| \cdot |B_V \cap N|/|N| = t\,|V|^k$. □

Lemma 3.4 (Maximality of stable product normalizers)**.** *If in addition* $[V]$ *is a maximal element of* $\mathcal{P}_X(S)$ *and* V *is normally saturating, then* B_V *is maximal in* G.

Proof. Let $B_V \le H \le G$. Since $H \supseteq B_V$ and $B_V N = G$ we have $HN = G$, so H surjects onto G/N and in particular acts coordinate-transitively.

Step 1 (the coordinate components of H recover X). Let H_1, G_1 be the coordinate-1 stabilizers. For $g \in G_1$ write $g = hn$ ($h \in H$, $n \in N$); n fixes every coordinate, hence so does h, so $h \in H_1$ and $\pi_1(G_1) = \pi_1(H_1) \cdot \pi_1(N \cap G_1) = \pi_1(H_1) \cdot \mathrm{Inn}(S)$. Therefore $\pi_1(H_1)$ and $\mathrm{Inn}(S)$ together generate X, and a class of subgroups of S is X-stable iff it is stable under $\pi_1(H_1)$.

Step 2 (coordinate kernels and saturation: $H \cap N$ is a product). Let $T := H \cap N \supseteq V^k$, let $W_i := \pi_i(T) \le S$ be the coordinate projections and $K_i := T \cap S_i$ the coordinate kernels (S_i the i-th factor). Then $V \le K_i$ (as $V^k \cap S_i = V$ in coordinate i), $K_i \trianglelefteq W_i$, and $V \le W_i$. Since K_i is a normal subgroup of W_i containing V, it contains $\langle V^{W_i} \rangle = W_i$ by saturation. So $K_i = W_i$ for every i and $T = \prod_i W_i$.

Step 3 (the W_i force $H = B_V$ or $H = G$). T is H-invariant. For $h \in H_1$, projecting $T^h = T$ to the first coordinate gives $W_1^{\pi_1(h)} = W_1$, so $[W_1]$ is $\pi_1(H_1)$-stable, hence X-stable by Step 1. For any i, coordinate-transitivity of H supplies $h \in H$ carrying coordinate i to coordinate 1 through a component $c \in X$, and projecting $T^h = T$ gives $W_1 = W_i^{\,c}$; applying X-stability of $[W_1]$ to c^{-1} gives $[W_i] = [W_1]$. Write $[W]$ for this common X-stable class. If $W = S$ then $T = N$, so $H \supseteq N$ and $H = HN = G$. Otherwise $1 < V \le W < S$. Form the normalizer tower $W \le N_S(W) \le N_S(N_S(W)) \le \cdots$; it is strictly increasing until it becomes self-normalizing, and it cannot reach S (the penultimate term would be a proper normal subgroup of the simple group S). Its terminal member U is proper, self-normalizing, and $[U]$ is X-stable ($N_S(W^a) = N_S(W)^a$, so stability propagates up the tower), i.e. $[U] \in \mathcal{P}_X(S)$ with $[U] \ge [W] \ge [V]$. Maximality of $[V]$ in $\mathcal{P}_X(S)$ forces $[U] = [V]$; then $|U| = |V|$ and $V \le W \le U$ gives $V = W = U$. Since $V \le W_i$ by Step 2 and $[W_i] = [W]$ gives $|W_i| = |W| = |V|$, each $W_i = V$; so $T = \prod_i W_i = V^k$, hence $|H| = |HN| \cdot |H \cap N|/|N| = t\,|V|^k = |B_V|$ and $H = B_V$. $\square$

Lemma 3.5 (Nonconjugacy of distinct stable classes)**.** *Distinct $[V] \ne [W]$ in $\mathcal{P}_X(S)$ give non-conjugate B_V, B_W.*

Proof. If $B_V^g = B_W$ then $(B_V \cap N)^g = B_W \cap N$ (as $N \trianglelefteq G$), i.e. $(V^k)^g = W^k$. Reading off any coordinate, W is the image of V under an element of X composed with a coordinate permutation, so $[W] = [V^a] = [V]$ for some $a \in X$ by X-stability, a contradiction. $\square$

The next lemma supplies, structurally, all hypotheses of Lemma 3.4 for the non-maximal flag-parabolic substitutes used in Appendix A: the standard parabolic subgroups whose maximal parabolic overgroups are fused by a graph automorphism. We use the notation of [6]. Here S is an untwisted simple group of Lie type with a split BN-pair of rank $\ell \ge 2$; all our applications are untwisted. Write $B = U \rtimes T$ for a Borel subgroup, with unipotent radical U and maximal torus T, and let Δ be the node set of the Dynkin diagram. For $J \subseteq \Delta$, let $Q_J \supseteq B$ be the standard parabolic whose Levi subgroup L_J is generated by T and the root subgroups corresponding to J. The maximal parabolic omitting node i is $P_i = Q_{\Delta \setminus \{i\}}$.

Lemma 3.6 (Flag-parabolic stability under graph fusion)**.** *Fix $J \subseteq \Delta$ and put $V := Q_J$. Suppose X is such that*

(a) *the S-class $[Q_J]$ is X-stable, and*
(b) *for every $J \subsetneq K \subsetneq \Delta$ the class $[Q_K]$ is not X-stable.*

Then $[V] \in \mathcal{P}_X(S)$, $[V]$ is a maximal element of $\mathcal{P}_X(S)$, and V is normally saturating. Consequently Lemmas 3.3–3.5 apply to V with no computational hypothesis.

Proof. Parabolic subgroups are self-normalizing [6, Thm. 8.3.3, p. 112], so $[V] \in \mathcal{P}_X(S)$ by (a).

Overgroups. Every subgroup containing V contains B, and every subgroup containing a Borel subgroup is a parabolic Q_K, $K \supseteq J$, of the same BN-pair [6, Thm. 8.3.2, p. 112]; moreover Q_K and $Q_{K'}$ are S-conjugate iff $K = K'$ [6, Thm. 8.3.3, p. 112].

Poset-maximality. If $[U] \in \mathcal{P}_X(S)$ with $V \le U^s < S$ for some s, then $U^s = Q_K$ for some $J \subseteq K \subsetneq \Delta$; $K = J$ gives $[U] = [V]$, while $J \subsetneq K$ contradicts (b), since $[U] = [Q_K]$ would be X-stable.

Saturation. Let $V \le W \le S$, so $W = Q_K$ ($K \supseteq J$) or $W = S$; we must show $\langle V^W \rangle = W$. The normal closure of V in W contains that of B. Write $W = U_K \rtimes L_K$ (Levi decomposition; for $W = S$ read $U_K = 1$, $L_K = S$). Then $U_K \le B$, and modulo U_K the image of B is a Borel subgroup $B_L = T \cdot (U \cap L_K)$ of L_K containing the full torus T. For each node $\alpha \in K$ the reflection representative n_α lies in L_K, and $U_{-\alpha} = U_\alpha^{n_\alpha}$. The subgroup $\langle B_L^{L_K} \rangle$ contains every L_K-conjugate of every subgroup of B_L; in particular it contains T, each U_α for $\alpha \in K$, and each $U_{-\alpha} = U_\alpha^{n_\alpha}$. These generate L_K by the root-subgroup description of the Levi decomposition [6, §8.5, pp. 118–120]. So $\langle B^W \rangle \supseteq U_K \cdot L_K = W$. □

Corollary 3.7 (Stable product-supplement engine). *Let $[V] \neq [W]$ be maximal elements of $\mathcal{P}_X(S)$ and suppose that both V and W are normally saturating. Then*

$$B_V = N_G(V^k), \qquad B_W = N_G(W^k)$$

are non-conjugate maximal supplements of N, with

$$B_V \cap N = V^k, \quad B_W \cap N = W^k, \quad |B_V| = t|V|^k, \quad |B_W| = t|W|^k.$$

Either or both of the selected classes may instead be represented by a flag parabolic satisfying the hypotheses of Lemma 3.6; the same conclusion then holds for the resulting pair.

Proof. Apply Lemmas 3.3–3.5; Lemma 3.6 supplies the stable-poset maximality and normal-saturation hypotheses in the flag case. □

The corollary isolates the exact output needed below: two non-conjugate maximal supplements and their orders. No family-specific classification data enters this engine.

4. The stable-class divisibility criterion

The purpose of this section is to convert the supplement engine into the uniform obstruction that drives the paper. The key comparison is between a missing p-part repeated in every socle coordinate and the smaller p-budget available in the wreath top.

For a prime p, v_p denotes the p-adic valuation and $s_p(k)$ the digit sum of k in base p, so that $v_p(k!) = (k - s_p(k))/(p-1)$.

Theorem 4.1 (Uniform stable-class divisibility criterion). *Let S be non-abelian simple, let $\mathrm{Inn}(S) \le X \le \mathrm{Aut}(S)$, and put $x := |X/\mathrm{Inn}(S)|$. Let $[V] \neq [W]$ be maximal elements of $\mathcal{P}_X(S)$, both normally saturating. If some prime p satisfies*

$$\eta = \eta_p(V, W) := v_p(|S|) - v_p(|V|) - v_p(|W|) \; > \; v_p(x),$$

then, for every $k \ge 2$, no finite group G with unique minimal normal subgroup $N \cong S^k$, whose action on the factors of N has coordinate closure X, has property P.

Proof. Conjugate the wreath realization as in Lemma 2.5; then Conventions 2.6–2.7 and Lemmas 3.3–3.5 apply. In particular, B_V and B_W are non-conjugate maximal subgroups of G, so property P forces $G = B_V B_W$ and

$$|B_V \cap B_W| = \frac{|B_V|\,|B_W|}{|G|} = t \cdot \Big(\frac{|V|\,|W|}{|S|}\Big)^k \in \mathbb{Z},$$

whence $p^{\eta k} \mid t$. But Lemma 2.8 gives

$$v_p(t) \;\le\; k\,v_p(x) + v_p(k!) \;=\; k\,v_p(x) + \frac{k - s_p(k)}{p-1} \;<\; k\big(v_p(x)+1\big) \;\le\; \eta k,$$

a contradiction. □

Example 4.2 (The criterion for A_5)**.** Let $S = A_5$. For every $\mathrm{Inn}(S) \le X \le \mathrm{Aut}(S)$, the classes of $V = A_4$ and $W \cong S_3$ are X-stable: they are the unique classes of maximal subgroups of orders 12 and 6, respectively. Thus they satisfy the structural hypotheses of Theorem 4.1. With $p = 5$,

$$\eta_5(V, W) = v_5(60) - v_5(12) - v_5(6) = 1.$$

Since $|X/\mathrm{Inn}(S)|$ divides $|\mathrm{Out}(A_5)| = 2$, its 5-valuation is zero. Theorem 4.1 therefore excludes every admissible group with socle A_5^k at once, for all $k \ge 2$. This small example displays the complete workflow: stable maximal classes, one obstruction prime, one strict gap, and no enumeration over k.

Remark 4.3. (i) The criterion is uniform in k, and this uniformity handles the infinite families. Subgroup sizes alone cannot yield a uniform contradiction: using $t \le x^k k!$, the natural upper bound for $|B_V||B_W|/|G|$ is

$$\left(x\frac{|V||W|}{|S|}\right)^k k!,$$

whose k-th root is $x|V||W|\,(k!)^{1/k}/|S|$ and therefore diverges. The valuation gap is essential. (ii) Only the *existence* of B_V, B_W is used; no classification of the maximal subgroups of G (or even of S) is needed. In particular the criterion is insensitive to completeness of tabulated maximal-subgroup data: it needs only that the two exhibited subgroups are maximal, or are flag-parabolic substitutes covered by Lemma 3.6, and that their classes are stable. (Machine *verification* of stability for the finite base does use enumerated subgroup data; see the trust boundary in Appendix C.)

A related monolithic-coordinate construction appears in the study of finite groups whose maximal subgroups have exact complements [30]. There the obstruction is not two-supplement integrality: a coordinate action is lifted to a primitive product action, and the absence of a regular subgroup excludes a complement. The wreath-top valuation budget is shared preliminary machinery for two otherwise different arguments; neither proof reduces to the other criterion.

The criterion has a $k = 1$ counterpart, which recovers the almost simple case.

Theorem 4.4 ($k = 1$: the almost simple case)**.** *Let $S \le G \le \mathrm{Aut}(S)$ with S non-abelian simple, and put $X := G$, so that $x := |G/S| = t$. Let $[V] \ne [W]$ be maximal elements of $\mathcal{P}_X(S)$, both normally saturating, and set $B_V := N_G(V)$, $B_W := N_G(W)$. Then:*

(1) *$B_V \cap S = V$, $B_V S = G$, and $|B_V| = t|V|$;*
(2) *B_V and B_W are non-conjugate maximal subgroups of G;*
(3) *if $\eta = v_p(|S|) - v_p(|V|) - v_p(|W|) > v_p(x)$ for some prime p, then G does not have property* P.

The proof is recorded in Appendix A. It is retained as a comparison with the $k \ge 2$ mechanism and is not used in the proof of the main theorem.

Convention 4.5. From now on, "S is *excluded*" means: for every X with $\mathrm{Inn}(S) \le X \le \mathrm{Aut}(S)$ there exist classes as in Theorem 4.1; consequently no finite group G with unique minimal normal subgroup S^k ($k \ge 2$, any admissible embedding) has property P. The almost simple branch $k = 1$ is supplied separately by Tikhonenko–Tyutyanov [34].

5. Two representative applications

The criterion is most useful when the family work is reduced to a repeatable sequence: select stable classes, prove maximality, choose an obstruction prime, check the group and subgroup valuations, bound the outer contribution, and route the exceptions. We retain one ordinary application and one graph-fusion application here. Appendix A gives the remaining family proofs in the same format.

5.1. **Ordinary stable maximal classes: alternating groups.**

Theorem 5.1. *$S = A_n$, $n \geq 5$, is excluded.*

Proof. Finite exceptions. For $n \leq 14$ this is covered by finite GAP certificates: the cases $n \leq 10$ fall under Proposition B.1 (including $n = 6$, whose exceptional outer automorphisms fuse classes; all five X are handled there), and the cases $11 \leq n \leq 14$ are covered by the finite certificates of Appendix B. Assume $n \geq 15$, so $\mathrm{Aut}(A_n) = S_n$ and $x \mid 2$.

Obstruction prime. Let p be the largest prime $\leq n$. We first record the bound $p \geq n/2 + 3$. By Nagura's theorem [27, Theorem, pp. 180–181], there is a prime in $(y, 6y/5]$ for $y \geq 25$. Thus, for $n \geq 31$, there is a prime in $(5n/6, n]$, and $5n/6 > n/2 + 3$ for $n > 18$. For $15 \leq n \leq 30$, the largest prime $p \leq n$ satisfies the bound by inspection: $p = 13, 13, 17, 17, 19, 19, 19, 19, 23, 23, 23, 23, 23, 23, 29, 29$ for $n = 15, \ldots, 30$. In particular, $n/2 < p \leq n$.

Selected classes, maximality, and stability. For $m < n/2$ let $M_m := (S_m \times S_{n-m}) \cap A_n$, and for n even let $I := (S_{n/2} \wr S_2) \cap A_n$. Each M_m is maximal in A_n, and I is maximal except in a finite list of cases, all with $n \leq 14$ [20, Thm. 1 and Tables I–II, pp. 366–368]. Conjugation by S_n preserves the partition shape (resp. block structure), each shape is a single A_n-class, and $\mathrm{Aut}(A_n) = S_n$ for $n \neq 6$: all these classes are X-stable for every X.

Occurrence, avoidance, and outer contribution. For any two distinct classes among $\{M_m : n - p < m < n/2\} \cup \{I \text{ if } n \text{ even and } n/2 < p\}$, we have $v_p(|A_n|) = 1$, while v_p of each subgroup order is 0 (all parts of the partitions involved are $< p$), and $v_p(x) \leq v_p(2) = 0$. Thus $\eta = 1 > 0$, and Theorem 4.1 applies, provided two such classes exist. The number of admissible intransitive m is $p - \lfloor n/2 \rfloor - 1$, plus one for I when n is even; two classes exist because $p \geq n/2 + 3$.

Conclusion. Theorem 4.1 excludes A_n for every $n \geq 15$, while the exact finite certificates cover the preceding degrees. ∎

The alternating family illustrates the ordinary mechanism: maximality and stability are visible from the permutation action, and a prime near n avoids two selected subgroup orders. Nothing in the argument depends on a fixed value of the socle multiplicity.

5.2. **Graph fusion: flag parabolics in projective linear groups.**

The next proposition exhibits the feature that ordinary examples do not show. A graph automorphism fuses opposite maximal-parabolic classes, so one replaces them by parabolics attached to invariant flags. Lemma 3.6 then recovers maximal supplements even though the coordinate subgroups themselves need not be maximal in S.

For $S = \mathrm{PSL}(n, q)$, the *graph part* of X means the image of X in the quotient of $\mathrm{Out}(S)$ by its diagonal and field subgroup. Thus X has nontrivial graph part precisely when this image contains the nontrivial A_{n-1} diagram involution. Appendix A.5 gives the corresponding definition for the other graph and graph–field families.

Proposition 5.2 (Projective flag-parabolic application)**.** *Let $S = \mathrm{PSL}(n, q)$ with $n \geq 4$, and let $\mathrm{Inn}(S) \leq X \leq \mathrm{Aut}(S)$ have nontrivial graph part. Then the hypotheses of Theorem 4.1 hold for this X. Consequently no admissible group with socle S^k, $k \geq 2$, and coordinate closure X has property* P.

Proof. Write $q = p^f$, $d = \gcd(n, q-1)$, and number the nodes of the A_{n-1} diagram in the usual order. The graph involution δ sends node i to $n-i$.

Selected classes. If $n = 4$, use the maximal parabolic P_2 and the flag parabolic $Q_{\{2\}}$, the stabilizer of an incident point–hyperplane pair. If $n \geq 5$, put

$$J_1 = \Delta \setminus \{1, n-1\}, \qquad J_2 = \Delta \setminus \{2, n-2\},$$

and use Q_{J_1} and Q_{J_2}, the stabilizers of an incident point–hyperplane flag and an incident line–$(n-2)$-space flag.

Maximality and stability. For $n = 4$, node 2 is fixed by δ, so $[P_2]$ is X-stable. The only proper parabolic overgroups of $Q_{\{2\}}$ are P_1 and P_3, which are exchanged by δ; hence Lemma 3.6 applies. For $n \geq 5$, both deleted node sets are δ-orbits. The proper parabolic overgroups of Q_{J_1} are P_1, P_{n-1}, and those of Q_{J_2} are P_2, P_{n-2}; each pair is exchanged by δ. Lemma 3.6 applies to both classes.

Obstruction prime and occurrence in $|S|$. If $n = 4$, let r be a primitive prime divisor of $p^{4f}-1$; then $r \mid \Phi_4(q) \mid |S|$. If $n \geq 5$, let r be a primitive prime divisor of $p^{nf} - 1$; then $r \mid \Phi_n(q)$, and this cyclotomic factor occurs in the order of S up to the central divisor d.

Avoidance of the selected subgroups. For $n = 4$, the relevant Levi exponents are at most $2f < 4f$. For $n \geq 5$, the Levi blocks have sizes $(1, n-2, 1)$ and $(2, n-4, 2)$, so their exponents are at most $(n-2)f$ and $\max(2, n-4)f$, both smaller than nf. Primitivity therefore gives $r \nmid |V||W|$ for the selected pair.

Outer contribution. Since $x = |X/\operatorname{Inn}(S)|$ divides $2df$ and $r \equiv 1 \pmod{nf}$ (with $n = 4$ in the first branch), we have $r \nmid 2f$; also $r \nmid d$, since $r \mid q-1$ would force $\operatorname{ord}_r(p) \mid f$. Hence $v_r(x) = 0 < v_r(|S|)$.

Exceptions. There is no positive-base Zsigmondy exception when $n = 4$. For $n \geq 5$, the only one is $p^{nf} = 2^6$, which gives $(n, f) = (6, 1)$. Take $r = 31$; the two flag Levi orders contain only $\mathrm{GL}_4(2)$- and $\mathrm{GL}_2(2)$-factors and 2-powers, while $v_{31}(|\operatorname{PSL}(6,2)|) = 1$ and $x \mid 2$.

For this fixed X, Theorem 4.1 now excludes every admissible group with socle S^k, $k \geq 2$, and coordinate closure X. □

This application isolates the graph-fusion yield: the stable object is an invariant flag, not an individual fused maximal-parabolic class. The stable-poset lemma makes that replacement available uniformly in every socle multiplicity.

6. Classification coverage and proof of the main theorem

The preceding sections contain the reduction, the reusable criterion, and two representative applications. Table 2 routes every remaining simple family to a detailed appendix proof or a finite certificate claim; it also identifies the common source of the obstruction prime and the exceptional route. Thus the logical path to Theorem 1.1 is visible without reading the repetitive calculations first.

Proposition 6.1 (Coverage)**.** *Every non-abelian finite simple group is excluded in the sense of Convention 4.5.*

Proof. By the classification of finite simple groups [12, Ch. 1, Table I, pp. 8–10] (see also [28, Thm. 1.1]), S is alternating, of Lie type, or one of the 26 sporadic groups. The cases are as follows. The simplicity exceptions are those in [13, Thm. 2.2.7(a)]; the low-rank classical identifications and the exceptional derived-group identifications are recorded in [8, §2.4, pp. xi–xii, and §3.5, p. xv] (with the individual Atlas entries pinpointed in the source map in [29]).

- A_n $(n \geq 5)$: Theorem 5.1.
- $\operatorname{PSL}(2, q)$ $(q \geq 4)$: Theorem A.1.

Family	Stable classes	Prime source	Exceptional handling	Detailed proof
Alternating	Intransitive and imprimitive maximals	Prime near n	Small degrees certified	§5
$\mathrm{PSL}(2,q)$	Borel or torus normalizers	Defining or primitive prime	Small fields certified	App. A.1
Lie type, no graph fusion	Two maximal parabolics	Zsigmondy	Finite base or substitute prime	App. A.2
Twisted rank at least two	Relative maximal parabolics	Zsigmondy	Listed small parameters	App. A.3
Twisted rank one	Borel and geometric maximal	Zsigmondy	Small fields certified	App. A.4
Graph-automorphism families	Fixed parabolics, stable flags, split-torus normalizers, or stable fixed-point/subfield classes	Family-specific primitive prime	Finite certificates or substitutes	§5, App. A.5
Sporadic and Tits	Certified stable maximal classes	Certificate prime	Exact finite records	App. B

TABLE 2. Coverage of the finite simple groups by the stable-class criterion.

- $\mathrm{PSL}(n,q)$ $(n \geq 3)$, $D_n(q)$ $(n \geq 4)$, $E_6(q)$: Theorem A.5.
- $\mathrm{PSp}(2n,q)$: $n \geq 3$, or $n = 2$ with q odd: Theorem A.2; $\mathrm{Sp}(4,2^f)$, $f \geq 2$: Theorem A.5 ($\mathrm{Sp}(4,2)$ is not simple).
- $\Omega(2n+1,q)$, q odd, $n \geq 3$: Theorem A.2 (for q even, $B_n(q) \cong C_n(q)$, already listed; $\Omega(5,q) \cong \mathrm{PSp}(4,q)$).
- $E_7(q)$, $E_8(q)$; $F_4(q)$, $p \neq 2$; $G_2(q)$, $p \neq 3$: Theorem A.2. $F_4(2^f)$ and $G_2(3^f)$: Theorem A.5. ($G_2(2)$ and ${}^2G_2(3)$ are not simple; their derived groups $\mathrm{PSU}(3,3)$ and $\mathrm{PSL}(2,8)$ are covered.)
- $\mathrm{PSU}(n,q)$: $n = 3$: Theorem A.4; $n \geq 4$: Theorem A.3.
- ${}^2D_n(q)$ $(n \geq 4)$, ${}^3D_4(q)$, ${}^2E_6(q)$, ${}^2F_4(2^f)$ (f odd ≥ 3): all four families are covered by Theorem A.3; the Tits group ${}^2F_4(2)'$ by Proposition B.2. (${}^2D_2$, ${}^2D_3$ are $\mathrm{PSL}(2,q^2)$, $\mathrm{PSU}(4,q)$, already listed.)
- $\mathrm{Sz}(2^f)$ (f odd ≥ 3), ${}^2G_2(3^f)$ (f odd ≥ 3): Theorem A.4.
- Sporadic groups: Proposition B.2.

Every finite simple group appears (repetitions from exceptional isomorphisms are harmless), so the proposition follows. □

Proof of Theorem 1.1. Suppose not, and let G have least order among the non-soluble groups with property P. By Proposition 2.3, G has a unique minimal normal subgroup $N = S^k$, S non-abelian simple, and after conjugation $N \leq G \leq X \wr S_k$ as in Convention 2.6. If $k = 1$, then G is almost simple, contrary to the theorem of Tikhonenko–Tyutyanov [34]. Hence $k \geq 2$. By Proposition 6.1 there are classes $[V] \neq [W]$, maximal in $\mathcal{P}_X(S)$ and normally saturating, and a prime p with $v_p(|S|) - v_p(|V|) - v_p(|W|) > v_p(x)$. By Theorem 4.1, G does not have property P, a contradiction. □

Remark 6.2. The argument proves the following stronger statement: for *every* finite group G with a unique minimal normal subgroup S^k (S non-abelian simple, $k \geq 2$), some pair of non-conjugate maximal subgroups of G fails to factorize G. The reduction of §2 is needed only to pass from an arbitrary non-soluble group to this configuration.

7. Conclusion

The transferable result is the stable-class divisibility criterion: two local classes with a strict valuation gap rule out every multiplicity k at once. Its subgroup hypotheses are structural rather than cosmetic. X-stability and self-normalization give exact coordinate intersections and orders, while maximality in $\mathcal{P}_X(S)$ and normal saturation are what promote the resulting normalizers to maximal supplements. Flag parabolics show that the method can survive automorphism fusion even when no individual maximal parabolic class is stable.

The global coverage still depends on CFSG and on published maximal-subgroup, order, automorphism, parabolic, and primitive-prime-divisor results. The finite and sporadic residue is computationally certified, and the formal developments verify only the conditional components stated in Appendix C; they do not replace those external inputs or give an end-to-end formal proof.

Li and Yang's subsequent product–socle lifting preprint [22] gives a structurally different route through the direct-power branch. The same wreath-top budget also appears in the exact-complement problem [30], but its final obstruction is different. Natural next questions are whether the supplement criterion applies to other prescribed products of maximal subgroups, whether its stable-poset hypotheses admit a classification-free source in broader families, and whether the remaining parabolic and finite-coverage interfaces can be formalized without importing the full classification ledger into a proof assistant.

Appendix A. Supplementary and infinite-family proofs

This appendix first records the unused $k = 1$ comparison and then discharges the repetitive infinite-family obligations while the main body retains the two mechanisms that teach the method. Every family case follows the same audit order: *selected classes*; *maximality and stability*; *obstruction prime*; *occurrence in* $|S|$; *avoidance of the selected subgroup orders*; *outer contribution*; *Zsigmondy or small-parameter exceptions*; and *conclusion by the stable-class criterion.* Closely related branches share a heading when the same argument proves several of these items at once.

Supplementary almost-simple argument.

Proof of Theorem 4.4. Supplement and order. By the Frattini argument via X-stability, as in Lemma 3.3, we have $B_V S = G$. Moreover, $B_V \cap S = N_S(V) = V$, and hence $|B_V| = t|V|$.

Maximality. The proof of Lemma 3.4 applies without its coordinate-kernel step; normal saturation enters only through that step and is not needed at $k = 1$ (the hypothesis is kept for uniformity with Theorem 4.1). Let $B_V \leq H \leq G$ and put $T := H \cap S$, an H-invariant subgroup containing V. If $T = S$, then $H \supseteq S$ and $H = HS = G$. Otherwise $G = B_V S = HS$. For $g = hs$, with $h \in H$ and $s \in S$, one has $T^g = (T^h)^s = T^s$; hence $[T]$ is G-stable and, since $X = G$, also X-stable. The normalizer tower of T ends at a proper self-normalizing subgroup U. Stability propagates up the tower, so $[U] \in \mathcal{P}_X(S)$ and $[U] \geq [V]$. Poset maximality of $[V]$ forces $[U] = [V]$, and $V \leq T \leq U$ gives $T = V$. Thus $H \leq N_G(V) = B_V$.

Nonconjugacy. If $B_V^g = B_W$ with $g \in G$, then $V^g = (B_V \cap S)^g = B_W \cap S = W$, and X-stability of $[V]$ gives $[W] = [V]$, a contradiction.

Arithmetic. Property P would force $G = B_V B_W$, so $t\,|V||W|/|S| \in \mathbb{Z}$, equivalently $p^\eta \mid t = x$, contrary to $\eta > v_p(x)$. ☐

Applying Theorem 4.4 to the witnesses in Proposition 6.1 gives a supplementary proof that no almost simple group has property P. This comparison is not used in the proof of Theorem 1.1, whose $k = 1$ branch continues to use Tikhonenko–Tyutyanov [34].

Common inputs for the family analysis. Three simplifications organize the ledger. First, when both selected classes consist of maximal subgroups of S, the stable-poset and normal-saturation hypotheses are automatic, so only stability and the valuation inequality need checking. Second, in the twisted families the automorphism structure induces no nontrivial permutation of the relevant relative-diagram types; diagonal and field automorphisms preserve the chosen parabolic classes. This includes the triality groups ${}^3D_4(q)$ and the Ree groups. Third, when a graph automorphism fuses maximal-parabolic classes, invariant flag parabolics become maximal stable classes by Lemma 3.6; this is the required flag-parabolic substitute for an individual fused maximal class.

Throughout this section S is simple, X is any group with $\mathrm{Inn}(S) \leq X \leq \mathrm{Aut}(S)$, and $x = |X/\mathrm{Inn}(S)|$. By Remark 3.2, when the two exhibited classes consist of maximal subgroups of S, only their X-stability and the valuation inequality require proof. We use Zsigmondy's theorem [40, specialized theorem, p. 283] throughout: for integers $a \geq 2$, $e \geq 3$ with $(a, e) \neq (2, 6)$, there is a prime r with $\mathrm{ord}_r(a) = e$ (a *primitive prime divisor* of $a^e - 1$); such r satisfies $r \equiv 1 \pmod{e}$. For $q = p^f$ we write $\Phi_e = \Phi_e(q)$ for cyclotomic values and always apply Zsigmondy to the prime p with exponent ef, so $\mathrm{ord}_r(p) = ef$ and $r \equiv 1 \pmod{ef}$. The source's additional positive-base exception has exponent 2, which lies outside every invocation here and therefore does not apply. An invocation-by-invocation audit of the hypotheses and exceptions is part of [29]. The $b = 1$ specialization and both exceptions are independently restated in [17, Thm. 2.2, p. 2].

The simple-group and outer-automorphism order formulas used below are those of [8, Tables 5–6, p. xvi]; for the classical families, the same formulas and their central divisors are also displayed in [8, §§2.1–2.4, pp. x–xii]. As an independent published derivation and tabulation we use [7, §10, pp. 220–221, and summary table, p. 239]. For parabolics, the statement that the unipotent radical is extended by the group obtained from the complementary subdiagram (with node-orbits in the twisted case) is [8, §3.6, p. xv]; the Levi decomposition and its root-subgroup description are [6, §8.5, pp. 118–120]. Thus the cyclotomic bounds below are not inferred from numeric tests: they come from the displayed group-order formulas applied to the named Levi types. The formula-to-source and parameter audit, including each exceptional substitute, is archived in [29]. In the exceptional case over $\mathbb{F}_2$ below, $\Omega^+(8, 2)$ and $P\Omega^+(8, 2)$ denote the same simple group; we use the shorter former notation when naming that finite group. For classical groups, argument notation such as $\mathrm{Sp}(4, 2^f)$ denotes a named simple group, whereas subscript notation such as $\mathrm{Sp}_2(q)$ and $\mathrm{GL}_2(q)$ is used for Levi factors.

A.1. **Projective special linear groups of rank one.**

Theorem A.1. *$S = \mathrm{PSL}(2, q)$, $q = p^f \geq 4$, is excluded.*

Proof. Finite exceptions and order data. For $q \in \{4, 5, 7, 8, 9, 11\}$ this is covered by the finite GAP certificates of Appendix B. All other simple parameters are covered uniformly below: odd $q \geq 13$ by Case A and even $q = 2^f$, $f \geq 3$, by Case B. Recall $|S| = q(q^2 - 1)/d$ with $d = \gcd(2, q-1)$, and $\mathrm{Out}(S) = C_d \times C_f$, so $x \mid df$: that is, $x \mid 2f$ for q odd and $x \mid f$ for q even.

Case A—selected classes, maximality, and stability. Suppose q is odd and $q \geq 13$. Let $U := N_S(T_+)$ and $V := N_S(T_-)$ be the normalizers of the split and nonsplit maximal tori. These are dihedral groups of orders $q - 1$ and $q + 1$, respectively. Both subgroups are maximal in S for $q \geq 13$ by Dickson's theorem [9, 16]; see also [1, Tables 8.1–8.2, p. 376, and Table 8.7, p. 380]. All split maximal tori are S-conjugate, as are all nonsplit maximal tori, and every automorphism preserves each type (there is no graph automorphism in type A_1; diagonal and field automorphisms act algebraically), so $[U]$ and $[V]$ are X-stable for every X.

Case A—prime, occurrence, avoidance, and outer contribution. Here the obstruction prime is the defining prime p and $|U||V|/|S| = d/q$, so $\eta = v_p(|S|) - 0 - 0 = f$, while $x \mid 2f$ gives $v_p(x) \leq v_p(f) \leq \log_p f < f = \eta$. Theorem 4.1 applies.

Case B—selected classes, maximality, and stability. Let $q = 2^f$ with $f \geq 3$. Here $|S| = q(q^2-1)$ and $x \mid f$. Let U be a Borel subgroup (the normalizer of a Sylow 2-subgroup), of order $q(q-1)$, and let $V := N_S(T_+)$ be dihedral of order $2(q-1)$. Both U and V are maximal for $q \geq 8$ [9, 16]; see also [1, Tables 8.1–8.2, p. 376]. $[U]$ is the class of Sylow-2 normalizers, and $[V]$ is the unique class of split torus normalizers; both are X-stable.

Case B—prime, occurrence, avoidance, outer contribution, and exception. Now $|U||V|/|S| = 2(q-1)/(q+1)$ with $q+1$ odd and coprime to $2(q-1)$, so it suffices to find a prime $r \mid q+1$ with $v_r(q+1) > v_r(f)$. If $f \neq 3$, take r a primitive prime divisor of $2^{2f}-1$: then $r \mid (2^{2f}-1)/(2^f-1) = q+1$ and $r \equiv 1 \pmod{2f}$, so $r > f$ and $v_r(x) = 0 < v_r(q+1) = \eta$. If $f = 3$, then $q = 8$ and $q+1 = 9$; taking $r = 3$ gives $\eta = 2 > 1 \geq v_3(x)$.

Conclusion. Theorem 4.1 applies in both parity branches, and the finite certificates route the remaining simple parameters. □

A.2. **Lie type without graph symmetry.**

Theorem A.2. *Let S be a simple group of Lie type of rank ≥ 2 over $\mathbb{F}_q$, $q = p^f$, of a type with no symmetry of its Dynkin diagram in the given characteristic:*

$$C_n = \mathrm{PSp}(2n,q)\ (n \geq 2,\ p \text{ odd if } n = 2), \quad B_n = \Omega(2n{+}1,q)\ (n \geq 3,\ q \text{ odd}),$$

$$G_2(q)\ (p \neq 3), \quad F_4(q)\ (p \neq 2), \quad E_7(q), \quad E_8(q).$$

Then S is excluded.

Proof. Torus exponent and selected classes. Write h for the relevant torus exponent: $h = 2n$ for B_n/C_n, and $h = 6, 12, 18, 30$ for G_2, F_4, E_7, E_8.

Let P_1, P_2 be maximal parabolic subgroups from two distinct classes (rank ≥ 2 guarantees these; take the ends of the diagram). They are maximal subgroups of S: any overgroup of a parabolic contains a Borel subgroup, and every subgroup containing a Borel is parabolic [6, Thm. 8.3.2, p. 112].

Maximality and automorphism stability. $\mathrm{Aut}(S)$ is generated by inner, diagonal, field and graph automorphisms [31], [4, Def. 3.1 and Thm. 3.4, pp. 31–32]. Diagonal and field automorphisms preserve types of vertices of the building, hence each parabolic class; type-permuting automorphisms arise only from diagram symmetries, absent for the listed types in the listed characteristics (the exceptional symmetries of B_2/C_2, F_4 in characteristic 2 and G_2 in characteristic 3 are excluded by hypothesis and treated in Theorem A.5). So every parabolic class is X-stable for every X.

Obstruction prime, occurrence, subgroup avoidance, and outer contribution. Let r be a primitive prime divisor of $p^{hf}-1$. Then: (i) $r \mid |S|$, since the order formula of each listed type contains the factor $q^h - 1$ (equivalently $\Phi_h(q)$) [8, Table 6, p. xvi]. (ii) r divides no maximal parabolic order: $|P_i| = q^{N_i}|L_i|(q-1)^{c_i}$ with L_i the semisimple part of the Levi factor, whose order is a product of a q-power and factors $q^j - 1$ with $j \leq h_i$, h_i the largest torus exponent of L_i; in every listed case every proper Levi has $h_i < h$. For B_n and C_n, the Levi types are respectively $A_{i-1}{\times}B_{n-i}$ and $A_{i-1}{\times}C_{n-i}$; in either case the relevant exponent is at most $\max(i, 2(n{-}i)) < 2n$. For the remaining types: G_2 has Levis A_1, $j \leq 2$; F_4: Levis $B_3, C_3, A_2{\times}A_1$, $j \leq 6$; E_7: Levis $E_6, D_6, A_6, \ldots$, exponents ≤ 12; E_8: Levis E_7, D_7, A_7, exponents ≤ 18. By primitivity r divides none of these, nor q, nor $q{-}1$. (iii) $r \nmid x$: $r \equiv 1 \pmod{hf}$ with $h \geq 4$, so $r > hf \geq 4f \geq d_S f \geq x$, where $d_S \leq \gcd(2, q-1)^2 \leq 4$ bounds the diagonal part and there is no graph part.

Thus $\eta = v_r(|S|) \geq 1 > 0 = v_r(x)$ for the pair $([P_1],[P_2])$ whenever r exists.

Zsigmondy and small-parameter exceptions. $p^{hf} = 2^6$ forces $p = 2$ and $hf = 6$: either $G_2(2)$ (not simple; $G_2(2)' \cong \mathrm{PSU}(3,3)$ is in the finite base) or $\mathrm{Sp}(6,2)$ (finite base). For $h \in \{12, 18, 30\}$, $hf = 6$ is impossible; B_n has q odd.

Conclusion. Theorem 4.1 applies to every nonexceptional parameter, and Appendix B supplies the two routed cases. □

A.3. **Twisted groups of twisted rank at least two.**

Theorem A.3. *Let S be one of*

$$\mathrm{PSU}(n,q)\ (n\geq 4),\quad P\Omega^-(2n,q)={}^2D_n(q)\ (n\geq 4),$$
$${}^3D_4(q),\quad {}^2E_6(q),\quad {}^2F_4(q)\ (q=2^f,\ f\ \mathit{odd}\geq 3).$$

Then S is excluded.

Proof. Selected classes and maximality. Each of these groups has a twisted BN-pair of rank at least 2, and therefore has at least two classes of maximal parabolic subgroups P_1, P_2. These subgroups are maximal in S, by the same argument as in Theorem A.2: every finite group of Lie type in defining characteristic has a split BN-pair [15, Example 1.16 and §1.3.5, pp. 10–11], and Carter's overgroup and self-normalizer theorems are stated for an arbitrary group with a BN-pair [6, Thms. 8.3.2–8.3.3, p. 112].

Automorphism stability. Parabolic preservation. Parabolic subgroups are precisely the overgroups of Borel subgroups, which are the normalizers of Sylow p-subgroups in defining characteristic [15, Example 1.16 and §1.3.5, pp. 10–11]. This description is automorphism-invariant. Hence every automorphism permutes the parabolic classes and acts on the relative type set.

Relative-type preservation. For a twisted group, the standard automorphism decomposition has no separate graph factor [4, Def. 3.3 and Thm. 3.4, p. 32]. The inner-diagonal and Frobenius representatives centralizing the defining Steinberg endomorphism preserve each orbit of simple roots, hence each node of the relative diagram. They normalize the standard Borel and torus, so they map every standard parabolic Q_J to a conjugate of the same relative type. Inner automorphisms fix classes. Consequently every parabolic class is X-stable for every X.

Rank-two diagnostic. In ${}^2F_4(q)$ the two maximal parabolic classes also cannot be interchanged because their Levi factors ${}^2B_2(q)$ and $\mathrm{SL}_2(q)$ are non-isomorphic [36, Thm. 4.5(i)–(ii), p. 166]. Diagram asymmetry alone would not suffice: the relative diagrams of $\mathrm{PSU}(4,q)$ and $\mathrm{PSU}(5,q)$ have a symmetric underlying graph, and the untwisted group $\mathrm{Sp}(4,2^f)$ admits a type-swapping automorphism.

Obstruction prime, occurrence in $|S|$, and subgroup avoidance. For each type, choose the exponent e and let r be a primitive prime divisor of p^e-1. In each case r divides $|S|$ but neither $|P_1|$ nor $|P_2|$:

- $\mathrm{PSU}(n,q)$, n odd: $e=2nf$, so $r\mid\Phi_{2n}(q)\mid q^n+1$, which divides $|\mathrm{SU}(n,q)|=q^{n(n-1)/2}\prod_{i=2}^n(q^i-(-1)^i)$. The Levi factor of P_i $(i=1,2)$ is a central product of $\mathrm{GL}_i(q^2)$ and $\mathrm{GU}(n-2i,q)$: its p'-order is a product of factors $q^{2j}-1$ $(j\leq 2)$ and $q^j-(-1)^j$ $(j\leq n-2)$, all dividing p^m-1 with $m\leq\max(4f,2(n-2)f)<2nf$. These order and Levi formulas are recorded in [8, §2.2, p. x, and §3.6, p. xv].
- $\mathrm{PSU}(n,q)$, n even: $e=2(n-1)f$, $r\mid\Phi_{2(n-1)}(q)\mid q^{n-1}+1$ (the $i=n-1$ factor). Levi exponents are $\leq\max(4f,2(n-3)f,(n-2)f)<2(n-1)f$ for $n\geq 4$ [8, §2.2, p. x, and §3.6, p. xv].
- ${}^2D_n(q)$, $n\geq 4$: $e=2nf$, $r\mid\Phi_{2n}(q)\mid q^n+1$, which divides $q^{n(n-1)}(q^n+1)\prod_{i=1}^{n-1}(q^{2i}-1)$. The Levi factor of P_i has type $\mathrm{GL}_i(q)\times O^-(2(n-i),q)$, and its relevant exponents are at most $\max(2f,2(n-1)f)<2nf$ [8, §2.4, p. xii, and §3.6, p. xv].
- ${}^3D_4(q)$: $e=12f$, $r\mid\Phi_{12}(q)=q^4-q^2+1$, which divides $q^8+q^4+1=(q^4+q^2+1)(q^4-q^2+1)$. Since $|{}^3D_4(q)|=q^{12}(q^8+q^4+1)(q^6-1)(q^2-1)$, we have $r\mid|S|$. The two Levi factors are $\mathrm{SL}_2(q^3)\cdot(q-1)$ and $\mathrm{SL}_2(q)\cdot(q^3-1)$ (mod centers): p'-orders divide $(q^6-1)(q-1)$ and $(q^2-1)(q^3-1)$, exponents $\leq 6f<12f$ [8, Table 6, p. xvi, and §3.6, p. xv].

- ${}^2E_6(q)$: $e = 18f$, $r \mid \Phi_{18}(q) \mid q^9+1$, which divides $q^{36}(q^{12}-1)(q^9+1)(q^8-1)(q^6-1)(q^5+1)(q^2-1)$. The four maximal parabolic classes correspond to the four τ-orbits of nodes of E_6 (τ the order-two symmetry): $\{2\}$, $\{4\}$, $\{1,6\}$, $\{3,5\}$; the complementary τ-stable subdiagrams are A_5, $A_2{\times}A_1{\times}A_2$, D_4, $A_2{\times}A_1{\times}A_1$ with induced twists, so the Levi derived groups are $\mathrm{SU}(6,q)$, $\mathrm{SL}_2(q){\times}\,\mathrm{SL}_3(q^2)$, ${}^2D_4(q)$-type, and $\mathrm{SL}_3(q){\times}\,\mathrm{SL}_2(q^2)$, with cyclotomic exponents $\leq 10f$, $6f$, $8f$, $4f$ respectively, all $< 18f$. Any two classes serve as P_1, P_2 [8, Table 6, p. xvi, and §3.6, p. xv]. This diagram derivation agrees with the published list $({}^2A_5(q), {}^2D_4(q), A_1(q){\times}A_2(q^2), A_1(q^2){\times}A_2(q))$ in [26, Lemma 6.8, p. 82].
- ${}^2F_4(q)$, f odd ≥ 3: $e = 12f$, $r \mid \Phi_{12}(q) \mid q^6+1$, which divides $|{}^2F_4(q)| = q^{12}(q^6+1)(q^4-1)(q^3+1)(q-1)$. The two Levi derived groups are ${}^2B_2(q) \cong \mathrm{Sz}(q)$ and $\mathrm{SL}_2(q)$ [36, Thm. 4.5(i)–(ii), p. 166] and [24, Main Theorem, pp. 52–53]: p'-orders divide $(q^2+1)(q-1)^2$ and $(q^2-1)(q-1)$, exponents $\leq 4f < 12f$. (${}^2F_4(2)$ is not simple; its derived group, the Tits group, is in the finite base.)

Outer contribution. In every case x divides $d_S \cdot cf$, where the diagonal order d_S divides $q+1$ (types ${}^2A$, ${}^2E_6$) or 4 (type ${}^2D$) or is 1 (${}^3D_4$, ${}^2F_4$), and the field part has order cf with $c \leq 3$ [8, Table 5, p. xvi] and [4, Def. 3.3 and Thm. 3.4, p. 32]. Since $r \equiv 1 \pmod e$ with $e \geq 6f$, r exceeds the field-part order. If $r \mid d_S$, then either $r \mid q+1$, which would give $\mathrm{ord}_r(p) \mid 2f < e$, or $r \leq 4$, contradicting $r > e \geq 6$. Hence $\eta = v_r(|S|) \geq 1 > 0 = v_r(x)$ whenever the primitive prime exists.

Zsigmondy and small-parameter exceptions. r fails to exist only if $p^e = 2^6$. Here $e \geq 6$ always, and $e = 6$ occurs only for $\mathrm{PSU}(4,q)$ with $f = 1$; then $p^6 = 2^6$ forces $S = \mathrm{PSU}(4,2) \cong \mathrm{PSp}(4,3)$ [5, Thm. 5.3, p. 61]; this group, of order 25920, is settled by the finite certificate of Appendix B. No other case reaches $p^e = 2^6$.

Conclusion. Theorem 4.1 applies to every uniform branch, and the sole exceptional simple group is routed to the exact finite record. □

A.4. **Twisted groups of twisted rank one.**

Theorem A.4. *Let S be one of*

$$\mathrm{PSU}(3,q)\ (q \geq 3), \quad \mathrm{Sz}(q) = {}^2B_2(q)\ (q = 2^f,\ f\ \textit{odd} \geq 3), \quad {}^2G_2(q)\ (q = 3^f,\ f\ \textit{odd} \geq 3).$$

Then S is excluded.

Proof. Common selected class and stability. Each group has a single class of parabolic subgroups, namely the Borel subgroups $B = N_S(Q)$, $Q \in \mathrm{Syl}_p(S)$; B is a *maximal* subgroup (rank-one BN-pair) and its class is X-stable for every X. We pair it with a second X-stable maximal class avoiding a Zsigmondy prime.

$\mathrm{PSU}(3,q)$—*second class, maximality, and stability.* For $q \geq 8$, $|S| = q^3(q^3+1)(q^2-1)/d$, $d = \gcd(3, q{+}1)$; $|B| = q^3(q^2{-}1)/d$. Let V be the stabilizer of a nonisotropic point of the unitary geometry. It has order $q(q-1)(q+1)^2/d$ and is maximal for $q \geq 3$, $q \neq 5$ [1, Table 8.5, p. 379] (class $\mathcal{C}_1$). These stabilizers form a single class, preserved by every semilinear automorphism.

$\mathrm{PSU}(3,q)$—*prime, occurrence, avoidance, and outer contribution.* Take r a primitive prime divisor of $p^{6f}-1$. It exists because the exceptional equation $p^{6f} = 2^6$ would force $q = 2$, for which S is not simple. Then $r \mid \Phi_6(q) = q^2-q+1$, which divides $(q^3+1)/(q+1)$, and hence $r \mid |S|$; $\mathrm{ord}_r(p) = 6f$ rules out r dividing $q(q^2-1)$ or $q(q-1)(q+1)^2$. Also $x \mid 2df$, and $r \equiv 1 \pmod{6f}$ gives $r > 2f$, while $r \mid d \mid q+1$ would force $\mathrm{ord}_r(p) \mid 2f$. Thus $\eta := v_r(|S|) \geq 1 > 0 = v_r(x)$.
$\mathrm{PSU}(3,q)$—*small parameters and conclusion.* For $q \in \{3,4,5,7\}$ (in particular the maximality exception $q = 5$), S is in the finite base. Theorem 4.1 handles every other parameter.

$\mathrm{Sz}(q)$—*selected class, maximality, and stability.* Here $|S| = q^2(q^2+1)(q-1)$; $|B| = q^2(q-1)$. Let $V := D_{2(q-1)}$ be the normalizer of a cyclic subgroup of order $q-1$. It is maximal by Suzuki's classification [32]; see also [1, Thm. 7.3.5, p. 367, and Table 8.16, p. 384]. These normalizers form a single class, stable under $\mathrm{Aut}(S) = S \rtimes C_f$.

$\mathrm{Sz}(q)$—*prime, occurrence, avoidance, outer contribution, and exceptions.* Let r be a primitive prime divisor of $2^{4f} - 1$. Such a prime exists because $4f \geq 12$. Then $r \mid \Phi_4(q) = q^2 + 1$, so $r \mid |S|$, $r \nmid |B|$, $r \nmid |V|$; $x \mid f$, and $r \equiv 1 \pmod{4f}$ gives $r > f$. Thus $v_r(|S|) \geq 1 > 0 = v_r(x)$, and Theorem 4.1 applies. The minimum exponent is $4f \geq 12$, so there is no Zsigmondy exception.

${}^2G_2(q)$—*selected class, maximality, and stability.* Here $|S| = q^3(q^3+1)(q-1)$; $|B| = q^3(q-1)$. Let $V := C_S(\iota) \cong \langle\iota\rangle \times \mathrm{PSL}(2, q)$ for an involution ι. The involutions form a single S-class, so $[V]$ is $\mathrm{Aut}(S)$-stable. Moreover, $|V| = q(q^2 - 1)$, and V is maximal [19, Thm. C, pp. 33–34].

${}^2G_2(q)$—*prime, occurrence, avoidance, outer contribution, and exceptions.* Let r be a primitive prime divisor of $3^{6f} - 1$. Such a prime exists because $6f \geq 18$. Then $r \mid \Phi_6(q) = q^2 - q + 1$, which divides $(q^3 + 1)/(q + 1)$, and hence $r \mid |S|$; $\mathrm{ord}_r(3) = 6f$ rules out $r \mid q^3(q - 1)$ and $r \mid q(q^2 - 1)$; $x \mid f < r$. So $v_r(|S|) \geq 1 > 0 = v_r(x)$. (${}^2G_2(3)$ is not simple; ${}^2G_2(3)' \cong \mathrm{PSL}(2, 8)$ is covered by Theorem A.1.) The exponent $6f \geq 18$ has no Zsigmondy exception, so Theorem 4.1 applies.

The three branches therefore satisfy the common eight-step template and exclude every twisted-rank-one simple group in the theorem. □

A.5. **Untwisted types with graph symmetry.**

Theorem A.5. *Let S be one of*

$$\mathrm{PSL}(n, q)\ (n \geq 3), \quad P\Omega^+(2n, q) = D_n(q)\ (n \geq 4), \quad E_6(q),$$
$$\mathrm{Sp}(4, q)\ (q = 2^f,\ f \geq 2), \quad F_4(q)\ (q = 2^f), \quad G_2(q)\ (q = 3^f).$$

Then S is excluded.

Proof. We keep the standard-parabolic notation Q_J of Lemma 3.6. The *graph part* of X is the image of X in the quotient of $\mathrm{Out}(S)$ by its inner-diagonal and field part: a subgroup of $C_2 = \langle\delta\rangle$ for $\mathrm{PSL}(n, q)$ $(n \geq 3)$, D_n $(n \geq 5)$, E_6; a subgroup of S_3 (triality) for D_4; while for $\mathrm{Sp}(4, 2^f)$, $F_4(2^f)$, $G_2(3^f)$, $\mathrm{Out}(S)$ is cyclic of order $2f$ generated by the exceptional graph-field automorphism ρ with $\rho^2 = \varphi_p$ [4, Def. 3.1(b), p. 31, and Thm. 3.4, p. 32], and "X has graph part" means the image of X in $\mathrm{Out}(S)$ is not contained in $\langle\rho^2\rangle$. An automorphism acts on the building permuting vertex types by a diagram symmetry; automorphisms with trivial graph part preserve every type. By Convention 2.6 we may normalize the graph part up to $\mathrm{Aut}(S)$-conjugacy; for D_4 this lets us assume a C_2 graph part is generated by the symmetry fixing nodes $1, 2$ and swapping $3, 4$.

The two main branches below follow the appendix template. In Case A, each bullet states the selected maximal parabolics and primitive prime, then checks occurrence, Levi avoidance, the outer bound, and any exception. In Case B, Table 3 presents those data uniformly; prose is retained for the exceptional mechanisms that require more than a table entry.

Case A—trivial graph part: selected classes and arithmetic. Every parabolic class is X-stable, and the argument of Theorem A.2 applies. The selected pairs and obstruction primes are given below; the $E_6(q)$ entry refers forward to a construction valid for every X.

- $\mathrm{PSL}(n, q)$*: pair $([P_1], [P_2])$, r a primitive prime divisor of $p^{nf} - 1$.* Then $r \mid \Phi_n(q) \mid q^n - 1$, and this cyclotomic factor divides $|S| \cdot d$, where $d = \gcd(n, q - 1)$. The Levi factors, of $\mathrm{GL}_{n-1}(q)$-type and $\mathrm{GL}_2 \times \mathrm{GL}_{n-2}$-type, have cyclotomic exponents at most $(n - 1)f < nf$. Moreover, $x \mid 2df$. The congruence $r \equiv 1 \pmod{nf}$ rules out $r \mid 2f$, since $r > nf \geq 3f > 2f$, and also rules out $r \mid d$, since that would imply $r \mid q - 1$ and hence $\mathrm{ord}_r(p) \mid f$. The Zsigmondy exceptions occur when $p^{nf} = 2^6$. If $(n, f) = (3, 2)$, then $S = \mathrm{PSL}(3, 4)$, which is in the finite base for all ten X. If $(n, f) = (6, 1)$, then

$S = \mathrm{PSL}(6,2)$; take instead $r = 31$ ($\mathrm{ord}_{31}(2) = 5$) and the pair $([P_2],[P_3])$. Their Levi types $\mathrm{GL}_2 \times \mathrm{GL}_4$ and $\mathrm{GL}_3 \times \mathrm{GL}_3$ have exponents at most $4 < 5$, so $31 \nmid |P_2||P_3|$, whereas $v_{31}(|\,\mathrm{PSL}(6,2)|) = 1$ and $x \mid 2$.

- $D_n(q)$, $n \geq 4$*: pair* $([P_1],[P_2])$ *(singular-point and singular-line stabilizers).* Let r be a primitive prime divisor of $p^{2(n-1)f} - 1$. Then $r \mid \Phi_{2n-2}(q)$, which divides $q^{n(n-1)}(q^n - 1)\prod_{i=1}^{n-1}(q^{2i} - 1)$. The Levi factors, of D_{n-1}-type and $A_1{\times}D_{n-2}$-type, have exponents at most $\max((n-1)f, 2(n-2)f) < 2(n-1)f$. Moreover, $x \mid 24f$, accounting for diagonal, field, and graph factors of orders at most 4, f, and 6, respectively. Since $r \equiv 1 \pmod{2(n-1)f}$ and $2(n-1) \geq 6$, we have $r > 6f$ and $r \nmid 24$. The only Zsigmondy exception is $p^{2(n-1)f} = 2^6$, which gives $(n,f) = (4,1)$ and $S = \Omega^+(8,2)$; Case B3 handles this group for every X.
- $E_6(q)$*.* The pair used in Case B4 is stable for every X.
- $\mathrm{Sp}(4,2^f)$*.* Take the pair $([P_1],[P_2])$, and let r be a primitive prime divisor of $2^{4f} - 1$. Such a prime exists because $4f \geq 8$. It divides $\Phi_4(q) = q^2 + 1$. The Levi factors $\mathrm{Sp}_2(q)$ and $\mathrm{GL}_2(q)$ have exponents at most $2f < 4f$; here $x \mid f$ and $r > 4f$.
- $F_4(2^f)$*.* Take the pair $([P_1],[P_2])$, and let r be a primitive prime divisor of $2^{12f} - 1$. It divides $\Phi_{12}(q)$. All proper Levi factors, of types B_3, C_3, $A_2{\times}A_1$, and $A_1{\times}A_2$, have exponents at most $6f < 12f$, and $x \mid f < r$. This completes the $p = 2$ case excluded from Theorem A.2.
- $G_2(3^f)$*.* Take the pair $([P_1],[P_2])$, and let r be a primitive prime divisor of $3^{6f} - 1$. It divides $\Phi_6(q)$. Levi factors of type A_1 have exponents at most $2f < 6f$, and $x \mid f < r$. This is the $p = 3$ case of G_2.

Case B—nontrivial graph part: stable-class mechanism. The flag-parabolic substitutes below are standard parabolics Q_J handled by Lemma 3.6. Their node sets are invariant under the relevant diagram symmetry, while their proper parabolic overgroups occur in nontrivial graph orbits. The lemma therefore supplies stable-poset maximality and normal saturation. Table 3 gives the complete branch roadmap; the prose following it is reserved for triality, invariant flags, and the exceptional graph–field fixed-point constructions.

Table 3: Nontrivial-graph branches: selected classes, arithmetic, and exception routing.

Branch	Stable classes and maximality mechanism	Obstruction prime and Levi avoidance	Outer contribution and exceptions
B1: $\mathrm{PSL}(3,q)$	The Borel $B = Q_\varnothing$ is a flag-parabolic substitute because P_1, P_2 are interchanged. The split-torus normalizer $N_T \cong ((q-1)^2/d).S_3$, $d = \gcd(3, q-1)$, is maximal for $q \geq 5$ and its class is automorphism-stable [25, 14]; see also [1, Table 8.3, p. 378].	Let r be a primitive prime divisor of $p^{3f} - 1$. Then $r \mid \Phi_3(q)$, $r \nmid |B|$, and $r \nmid |N_T| = 6(q-1)^2/d$ because $r \equiv 1 \pmod 3$ gives $r \geq 7$.	$x \mid 2df$, while $r > 3f$ and $r \nmid d$. The cases $q \leq 4$ are in the finite range; the 2^6 exception is $\mathrm{PSL}(3,4)$.
B2: $\mathrm{PSL}(n,q)$, $n \geq 4$	For $n = 4$, use $[P_2]$ and $[Q_{\{2\}}]$; for $n \geq 5$, use the two invariant flag parabolics deleting $\{1, n-1\}$ and $\{2, n-2\}$. Proposition 5.2 proves stability and maximal-supplement status.	Use a primitive divisor of $p^{4f} - 1$ when $n = 4$, and of $p^{nf} - 1$ otherwise. The Levi block sizes are at most 2, $(n-2)$, and $\max(2, n-4)$, all below the primitive exponent.	$x \mid 2df$ and the primitive prime avoids $2df$. The sole exception is $\mathrm{PSL}(6,2)$, where $r = 31$ and the same flag pair applies.

Continued on next page

Table 3 (continued)

Branch	Stable classes and maximality mechanism	Obstruction prime and Levi avoidance	Outer contribution and exceptions
B3: $D_n(q)$	For $n \geq 5$, $[P_1], [P_2]$ are fixed by the graph involution. For D_4, the same pair handles trivial or normalized C_2 graph part; triality uses $[P_2]$ and $[Q_{\{2\}}]$. The triality orbit argument is given below.	Use a primitive divisor of $p^{2(n-1)f} - 1$; for D_4 the exponent is $6f$. The relevant Levi exponents are strictly smaller; triality gives A_1^3 and A_1.	$x \mid 24f$ and the primitive prime avoids this factor. The 2^6 exception $\Omega^+(8, 2)$ uses $r = 5$; the two graph branches are checked below.
B4: $E_6(q)$	The diagram involution fixes nodes $2, 4$, so $[P_2], [P_4]$ are stable maximal classes for every X.	Let r be a primitive prime divisor of $p^{12f} - 1$. Then $r \mid \Phi_{12}(q)$ occurs in $\lvert S\rvert$. Here the Levi of P_2 has type A_5 with exponents at most $6f$, and the P_4 Levi has type $A_2{\times}A_1{\times}A_2$ with exponents at most $3f$ [6, §8.5, pp. 118–120].	With $d = \gcd(3, q - 1)$, $x \mid 2df \leq 6f < r$. There is no Zsigmondy exception.
B5: $\mathrm{Sp}(4, 2^f)$	Use the Borel flag-parabolic substitute and $V = \mathrm{Fix}(\widetilde{\rho}^{\,f}) \cap S$: $V = \mathrm{Sz}(q)$ for odd f, and $V = \mathrm{Sp}(4, 2^{f/2})$ for even f. Maximality and stability are proved below.	For odd f, use a primitive divisor of $2^{2f} - 1$; for even f, one of $2^{4f} - 1$. The fixed-point subgroup and Borel avoid it.	$x \mid 2f$. The nonsimple $f = 1$ case routes to A_6; $q = 8$ uses the substitute prime 3. The finite $q = 4$ record checks the same construction.
B6: $F_4(2^f)$	Use the invariant flag parabolics $Q_{\{2,3\}}$ and $Q_{\{1,4\}}$; each pair of proper parabolic overgroups is interchanged by ρ. Details appear below.	A primitive prime divisor of $2^{12f} - 1$ avoids both selected classes. Their Levi types are B_2 and $A_1{\times}A_1$, with exponents at most $4f$ and $2f$ [6, §8.5, pp. 118–120].	$x \mid 2f < 12f < r$. There are no exceptions.
B7: $G_2(3^f)$	Use the Borel flag-parabolic substitute and $V = \mathrm{Fix}(\widetilde{\rho}^{\,f}) \cap S$: $V = {}^2G_2(q)$ for odd f, and $V = G_2(3^{f/2})$ for even f. Maximality and stability are proved below.	For odd f, use a primitive divisor of $3^{3f} - 1$; for even f, one of $3^{6f} - 1$. The factors $q^6 - 1$ and $q^2 - 1$ in the group order supply the required primes.	$x \mid 2f$. There is no Zsigmondy exception; $G_2(3)$ also has a finite certificate for both coordinate closures.

The table is a roadmap rather than a replacement for the exceptional mechanisms. We now justify the three points that are not immediate from the standard maximal-parabolic calculation.

Triality detail (B3). For $n \geq 5$, the graph part is at most the C_2 swapping the two spinor nodes $n{-}1, n$ and fixing $1, \ldots, n{-}2$, so the Case A pair $([P_1], [P_2])$ is stable for every X. For $n = 4$, the same statement holds for trivial or normalized C_2 graph part. If the graph part contains triality, use $([P_2], [Q_{\{2\}}])$. Node 2 is fixed by all of S_3, so $[P_2]$ is a stable maximal class with Levi type $A_1{\times}A_1{\times}A_1$. The flag parabolic $Q_{\{2\}}$ has Levi type A_1; its proper overgroups $Q_{\{1,2\}}, Q_{\{2,3\}}, Q_{\{2,4\}}$ and P_4, P_3, P_1 form two regular triality orbits on the end nodes $\{1, 3, 4\}$. None is X-stable, so Lemma 3.6 applies.

Let r be a primitive prime divisor of $p^{6f} - 1$. It avoids both classes, $v_r(\lvert S\rvert) \geq 1$, and $r \nmid x$ as in Case A. If $p = 2$ and $f = 1$, then $S = \Omega^+(8, 2)$ and $x \mid 24$. Take $r = 5 = \Phi_4(2)$. Then $v_5(\lvert S\rvert) = v_5\big(2^{12}(2^6 - 1)(2^4 - 1)^2(2^2 - 1)\big) = 2$ and $v_5(x) = 0$. For triality, both selected Levi orders are $\{2, 3\}$-numbers, so the gap is 2. For trivial or C_2 graph part, $v_5(\lvert P_1\rvert) = 1$ and $v_5(\lvert P_2\rvert) = 0$, so the gap is 1.

Exceptional graph–field fixed points (B5). Let $S = \mathrm{Sp}(4, 2^f)$ with $f \geq 2$ and let B be a Borel subgroup. Its two proper maximal-parabolic overgroups are interchanged by ρ, so Lemma 3.6 makes $[B]$ a stable-poset maximal class. Put

$$V := \mathrm{Fix}(\widetilde{\rho}^{\,f}) \cap S,$$

where $\widetilde{\rho}^2 = \varphi_2$ and $\widetilde{\rho}^{2f} = F_q$. Thus $V = \mathrm{Sz}(q)$ for odd f, whereas for even f it is the subfield subgroup $\mathrm{Sp}(4, 2^{f/2})$. Both are maximal [1, Table 8.14, p. 384] and [32]. Since $\widetilde{\rho}$ commutes with its own power, $\rho(V) = V$; inner automorphisms preserve $[V]$ by conjugation. Hence $[V]$ is $\mathrm{Aut}(S)$-stable.

Case f odd, $2f \neq 6$. Let r be a primitive prime divisor of $2^{2f} - 1$. It divides $q + 1$, so $v_r(|S|) = 2v_r(q + 1) \geq 2$, while $v_r(|B|) = v_r(|\mathrm{Sz}(q)|) = 0$ and $x \mid 2f < r$. If $2f = 6$, take $r = 3$; then $v_3(|\mathrm{Sp}(4, 8)|) = v_3(63 \cdot 4095) = 4 > 1 \geq v_3(x)$, while both selected subgroups have zero 3-valuation.

Case f even. Let r be a primitive prime divisor of $2^{4f} - 1$. Writing $q_0^2 = q$, one has $|\mathrm{Sp}(4, q_0)| = q_0^4(q_0^2 - 1)(q_0^4 - 1)$, whose cyclotomic exponents are at most $2f < 4f$. Hence r avoids both V and B, while it divides $|S|$ and $x \mid 2f < r$. The finite $\mathrm{Sp}(4, 4)$ certificate records this same construction with the primes 5 and 17.

Invariant flag parabolics (B6). For $S = F_4(2^f)$, the exceptional graph–field automorphism reverses $1-2 \Rightarrow 3-4$. The node sets $J_1 = \{2, 3\}$ and $J_2 = \{1, 4\}$ are invariant. The proper overgroups of Q_{J_1} are P_4 and P_1, which are interchanged; those of Q_{J_2} are $Q_{\{1,2,4\}}$ and $Q_{\{1,3,4\}}$, also interchanged and of distinct types. Lemma 3.6 therefore applies to both. The corresponding Levi types are B_2 and $A_1 \times A_1$, with exponents at most $4f$ and $2f$. Let r be a primitive prime divisor of $2^{12f} - 1$. It avoids both classes, $v_r(|S|) \geq 1$, and $x \mid 2f < 12f < r$; there is no exception.

Exceptional graph–field fixed points (B7). Let $S = G_2(3^f)$. The automorphism ρ interchanges the two maximal parabolics and satisfies $\rho^2 = \varphi_3$, so the Borel class is a flag-parabolic substitute. With $V := \mathrm{Fix}(\widetilde{\rho}^{\,f}) \cap S$, one has $V = {}^2G_2(q)$ for odd f and $V = G_2(3^{f/2})$ for even f. These subgroups are maximal [19, Thm. A, p. 33]; for $f = 1$, the identification ${}^2G_2(3) \cong \mathrm{P\Gamma L}_2(8)$ and its maximality in $G_2(3)$ are also recorded in the Atlas [8]. The same commuting-endomorphism argument as in B5 makes $[V]$ automorphism-stable.

Case f odd. Let r be a primitive prime divisor of $3^{3f} - 1$. It divides $\Phi_3(q) = q^2 + q + 1$, and $|S| = q^6(q^6 - 1)(q^2 - 1)$ shows that $r \mid |S|$. Primitivity gives $r \nmid |B| = q^6(q - 1)^2$ and $r \nmid |{}^2G_2(q)| = q^3(q^3 + 1)(q - 1)$; moreover $r \equiv 1 \pmod{3f}$ gives $r > 2f \geq x$. There is no Zsigmondy exception because the base is 3.

Case f even. Let r be a primitive prime divisor of $3^{6f} - 1$. Then $r \mid \Phi_6(q) = q^2 - q + 1$, while $r \nmid |B|$ and $r \nmid |G_2(3^{f/2})| = q^3(q^3 - 1)(q - 1)$; also $x \mid 2f < 6f < r$. The group $G_2(3)$ additionally carries a finite certificate for both X.

Conclusion. In every case the selected classes and prime satisfy Theorem 4.1, which excludes S^k for every $k \geq 2$. □

Appendix B. Finite and sporadic certificate claims

This appendix states exactly what the finite computations contribute. Command lines, package installation, certificate filenames, hashes, and mutation-test mechanics are kept in the repository supplement [29]; none of those operational details substitutes for the mathematical claims below.

B.1. **Finite-range coverage.**

Proposition B.1 (Finite-range coverage)**.** *Every non-abelian simple group S with $|S| \leq 1.05 \times 10^7$ is excluded in the sense of Convention 4.5.*

Proof. Inventory and routing. The certified inventory is the disjoint union of the 47 non-abelian simple groups with $|S| < 5 \times 10^5$ and the 51 groups with $5 \times 10^5 \leq |S| \leq 1.05 \times 10^7$. The committed replay regenerates the upper range with GAP's simple-group iterator. The lower range is read from its frozen canonical inventory and checked for certificate coverage and routing; the committed replay does not independently regenerate that lower range. In the upper inventory, the 38 groups $\mathrm{PSL}(2, q)$ are routed to the uniform proof in Appendix A.1; the remaining 13 receive finite certificates. Exactly two of those 13 are not excluded by ordinary maximal-class pairs: $\mathrm{Sp}(4, 4)$ for $x = 4$ and $\mathrm{PSL}(5, 2)$ for $x = 2$. Both are routed to the stable-poset substitute certificates described below.

The exception manifest has precisely seven simple groups above the displayed order bound: $A_{11}, A_{12}, A_{13}, A_{14}$ receive finite certificates used in Theorem 5.1, while $\mathrm{PSL}(6, 2)$, $\Omega^+(8, 2)$, and $\mathrm{Sp}(4, 8)$ are treated in Appendix A.5 by the stated substitute primes. Thus every finite or exceptional route that appeared in the original body-level inventory remains explicit in an appendix.

Ordinary and stable-poset substitute records. The repository retains the historical machine field `kind=novelty`; in the manuscript a *stable-poset substitute* means a selected class that need not be maximal in S but whose product normalizer is maximal in G by Lemma 3.4. For every certified simple group S and every coordinate closure $\mathrm{Inn}(S) \leq X \leq \mathrm{Aut}(S)$, a successful record identifies two distinct X-stable subgroup classes, their exact class identities and orders, an obstruction prime, and the strict valuation gap required by Theorem 4.1. For ordinary records the selected subgroups are maximal in S. A maximal-class pair is available for every coordinate closure except in designated closures of six simple groups:

$$\mathrm{PSL}(3, 2),\ \mathrm{PSL}(2, 11),\ A_6,\ \mathrm{PSL}(3, 4),\ \mathrm{Sp}(4, 4),\ \mathrm{PSL}(5, 2).$$

More precisely, the exceptional closures are $x = 2$ for $\mathrm{PSL}(3, 2)$ and $\mathrm{PSL}(2, 11)$; $X/\mathrm{Inn}(S) \in \{2_2, 2_3, 2^2\}$ for A_6; several recorded closures for $\mathrm{PSL}(3, 4)$; $x = 4$ for $\mathrm{Sp}(4, 4)$; and $x = 2$ for $\mathrm{PSL}(5, 2)$. For the first four groups, the computation additionally certifies normal saturation by three distinct overgroup and embedding-orbit implementations. They share the same GAP runtime, group representations, and pinned subgroup data, so their agreement is a cross-check within that shared trust base rather than independent software verification. For $\mathrm{Sp}(4, 4)$ and $\mathrm{PSL}(5, 2)$, Lemma 3.6 proves saturation and stable-poset maximality structurally. Thus Corollary 3.7 and Theorem 4.1 apply to every record.

Representative certificates. Example 4.2 records the ordinary tuple $(|A_4|, |S_3|, 5, 1)$. A stable-poset substitute record for $\mathrm{PSL}(3, 2)$ with $x = 2$ uses the pair $(S_3, 7{:}3)$ at $r = 2$, with valuation gap $2 > 1$. For $\mathrm{Sp}(4, 4)$ with $x = 4$, the pair is a Borel subgroup and the subfield subgroup $\mathrm{Sp}(4, 2)$, with certificate primes 5 and 17. For $\mathrm{PSL}(5, 2)$ with $x = 2$, the selected classes are the incident-flag parabolics $(Q_{\{2,3\}}, Q_{\{1,4\}})$, with certificate primes 5 and 31. The last two pairs are also instances of the uniform graph-fusion constructions in Appendix A.5.

Identity, completeness, and cross-checks. Each coordinate closure has a SHA-256 action fingerprint together with intrinsic quotient and kernel data; the checker requires every required closure exactly once. Each subgroup record carries its class identity, so same-order non-conjugate classes are not identified by order alone. The completeness assumptions are explicit: the two finite inventories and the relevant subgroup classes in the pinned GAP data are assumed complete. An independent routing and coverage check verifies that every inventory entry is routed exactly once, and the three saturation implementations fail on any disagreement. The repository supplement exposes the challenge data and the independently parsed successful records. □

B.2. **Sporadic groups and the Tits group.**

Proposition B.2 (Certified sporadic and Tits cases)**.** *Every sporadic simple group is excluded, as is the Tits group* ${}^2F_4(2)'$.

Proof. Inventory and certificate content. The six groups $M_{11}, M_{12}, M_{22}, M_{23}, J_1, J_2$ lie in Proposition B.1. The remaining 20 sporadic groups are

$$M_{24}, J_3, J_4, \mathrm{HS}, \mathrm{McL}, \mathrm{Co}_1, \mathrm{Co}_2, \mathrm{Co}_3, \mathrm{Suz}, \mathrm{He}, \mathrm{Ru}, \mathrm{O'N},$$
$$\mathrm{Fi}_{22}, \mathrm{Fi}_{23}, \mathrm{Fi}_{24}', \mathrm{HN}, \mathrm{Ly}, \mathrm{Th}, B, M.$$

For each of these groups and the Tits group, a certificate records two stable maximal-subgroup classes, their published class identities and orders, an obstruction prime, and the valuation gap. The resulting 42 selected-class records are bound to the group, character-table identifier, pinned maximal-list position, and exact order. If $\mathrm{Out}(S) = 1$, stability is automatic. If $|\mathrm{Out}(S)| = 2$, the selected order occurs exactly once in the complete pinned list of maximal-subgroup classes, so automorphisms must preserve that class. This unique-order argument is the only point here at which completeness of the pinned maximal-subgroup list is logically required; no stored class fusion is assumed.

Representative certificates. The Monster record uses maximal-list positions 45, 46, and the Baby Monster record uses positions 29, 30. Their respective pairs and primes are

$$(59{:}29, 41{:}40), \quad r = 71; \qquad (L_2(11).2, 47{:}23), \quad r = 31.$$

These examples are illustrative; the claim uses all 42 records rather than the samples alone.

The selected records are cross-checked against the published Atlas and pinned character-table-library data [2, 8], Wilson's corrected sporadic tables [37, §4, pp. 66–68], and their provenance account [3, §2, pp. 23–24]. The Tits entries also have the independent sources [35, pp. 553–563] and [33]; the Monster entries use Theorems 1.1–1.2 and Table 1 of [10, pp. 862–863]. These source checks identify the mathematical maximal-subgroup claims; GAP [11] supplies the finite certificates. □

B.3. **Family-arithmetic certificates.** The family proofs use an exact 34-branch arithmetic manifest. A universal checker matches every branch to its cited group order, Levi order, outer-automorphism bound, maximality source, and Zsigmondy invocation; it also routes all exceptional parameters to substitute primes or finite records. The positive-base Zsigmondy exception occurs in 12 branches, representing seven distinct exception records, and every one is matched to its finite destination or substitute-prime argument. A separately authored symbolic implementation reproduces all 34 branches, bounds, residue arguments, and exceptional valuations without reading that manifest, and Lean checks the resulting conditional arithmetic statements. The published formulas and classification results remain external inputs.

Finite sweeps over 1272 projective-linear prime powers, alternating degrees through 10^4, and 7892 Lie-type parameter instances, with ranks through 25 and field sizes through 3000, returned no failures. These are regression checks, not proofs of the universally quantified family statements. Their role is to detect transcription or implementation errors in the formulas proved in Appendix A, not to replace those proofs.

Appendix C. Verification and trust boundary

The proof spine and the classification/computation layer are separated in Tables 4 and 5, so that each row states both its evidentiary mode and its external assumptions.

Lean's strongest reader-facing output is the conditional product-supplement spine: exact subgroup orders, maximality, nonconjugacy, the wreath-top divisor, and a strict valuation gap imply that property P fails. The working source tree contains a Comparator package that states this proposition independently of the project implementation and checks the project theorem against that statement. It does not import CFSG coverage, the flag-parabolic substitute step, or finite GAP certificates as formal theorems.

Claim	Ordinary proof relative to cited inputs	Formal coverage	External or trusted input
Property P and quotient inheritance	Complete	Lean: complete	Standard finite-group definitions
Minimal-counterexample core	Complete	Lean: complete; Rocq interface input	Standard finite-group facts
Characteristically simple direct-power decomposition	Complete	Rocq producer; Lean reindexing consumer	Rocq–Lean semantic correspondence
Faithful wreath realization and quotient divisor	Complete	Lean: complete	Explicit direct-power equivalence
Product-supplement intersection and exact order	Complete	Lean: conditional	Stable-class hypotheses
Product-normalizer maximality	Complete	Lean: conditional	Stable-poset and saturation inputs
Nonconjugacy and valuation contradiction	Complete	Lean: conditional; Python arithmetic cross-check	Stable-class and order hypotheses
Main theorem end to end	Complete relative to cited external inputs	Not end-to-end formalized	CFSG, classifications, formal interface, and finite certificates

TABLE 4. Ordinary and formal status of the conceptual proof spine.

Coverage claim	GAP/Python role	Formal status	External or shared trust base
Universal family arithmetic	Manifest, symbolic implementation, and regression sweeps	Lean: conditional branch theorems	Published group, Levi, outer, maximality, and Zsigmondy data
Graph-fusion flag-parabolic substitutes	Selected finite checks only	`PAR-NOVELTY` not closed in Lean	Cited BN-pair and parabolic theory
Finite-range and sporadic coverage	GAP certificates and Python parsing	Not formalized	Pinned GAP, ATLAS, and character-table completeness
Cross-kernel coordinate interface	Signature and source-token checks	Rocq producer and Lean consumer each kernel-checked	Semantic correspondence between the two libraries
Evidence closure and mutation tests	Hash closure and deliberate fault detection	Not a mathematical proof layer	Shared files, runtimes, and pinned data are explicit

TABLE 5. Computational, classification, and cross-kernel trust boundary.

More explicitly, Lean checks quotient inheritance and the minimal-order core: uniqueness and characteristic simplicity of the minimal normal subgroup, its non-solubility, the solubility of the quotient, its trivial centralizer, and faithfulness of the conjugation action. Under the normalized coordinate hypotheses it checks the product-supplement intersection and exact order, maximality, nonconjugacy, the subgroup-product lower-divisibility bridge, and the universal-in-k arithmetic contradiction. It also checks the conditional arithmetic deductions for all 34 family branches. The group and Levi orders, outer-automorphism and maximality data, and Zsigmondy inputs remain the cited external hypotheses of those branch theorems.

The coverage manifest retains the historical identifier `PAR-NOVELTY`. It denotes the flag-parabolic substitute used when graph automorphisms fuse parabolic classes. It remains explicitly not closed in Lean; its proof is the ordinary argument of Lemma 3.6 and the representative application in §5, using the cited parabolic theory.

Rocq/MathComp proves that the relevant finite characteristically simple non-soluble group is an internal direct product of $k > 0$ automorphic copies of one non-soluble, non-abelian simple subgroup S, with $|N| = |S|^k$, and also constructs an isomorphism to the corresponding external coordinate product. Lean reindexes the nonempty coordinate type to $\{1, \ldots, k\}$ and consumes the resulting equivalence $N \cong S^k$. It then constructs the coordinate factors, proves that automorphisms permute them, constructs the faithful map $\mathrm{Aut}(S^k) \to \mathrm{Aut}(S) \wr S_k$, proves transitivity of the G-action on the factors, identifies the inverse image of $\mathrm{Inn}(S)^k$ as N, normalizes the coordinate closure, and proves

$$|G/N| \mid |X/\operatorname{Inn}(S)|^k k!.$$

No single kernel checks the translation between the two libraries' notions of subgroup isomorphism and external product; that translation is the stated trusted interface. The fail-closed formal interface checker identifies the exact producer, reindexer, and consumer declarations and rejects drift in the enumerated signatures and definition-source tokens. It does not decide arbitrary semantic equivalence across the two libraries. The mathematical identification between the two formulations therefore remains an audited trusted correspondence, along with both kernels and their pinned libraries.

GAP certifies only the finite inventories and subgroup-class facts stated in Appendix B. Python independently parses those certificates, checks coverage and source-map topology, and reproduces the family arithmetic. Finite parameter sweeps are regression tests, not proofs of universal statements. The proof-essential scripts are deterministic and fail closed: a skipped case, soft failure, ambiguous identity, missing datum, version mismatch, or byte-level certificate mismatch is fatal. Finite records retain class positions and guard fingerprints as well as orders, primes, and valuations; superseded exploratory scripts are excluded from the proof-evidence manifests. The main theorem therefore remains an ordinary mathematical proof supported by partial formal verification, certified finite computation, and explicit external classification inputs.

Data availability. The exact source and evidence tree is archived as release 1.1.1 [29]. Its GitHub tag and Zenodo version record contain the manuscript, formal developments, GAP programs, finite certificates, arithmetic manifests, source maps, independent checkers, and computational-environment recipe described here. The repository guide maps each computational claim to its artifact; Appendices B and C state the mathematical claims and trust boundary without requiring operational log names in the proof narrative. Runtime tools and direct downloads are version- or hash-pinned, but the Debian package snapshot and complete opam solver closure are not frozen; long-term replay is not fully hermetic at those layers.

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